

1 Module d'Entrée Générique

```
module Entry(sensor_id : integer);
input SENSOR_TRIGGER, OPEN, CLOSE, TICK;
output PERSON_ENTERED;
integer count := 0;
const integer K = 10; // P riode pour laisser entrer une personne
integer timer := 0;

loop
  await SENSOR_TRIGGER;
  count := count + 1;
  every TICK when count > 0 and present(OPEN) do
    if timer == K then
      emit PERSON_ENTERED;
      count := count - 1;
      timer := 0;
    else
      timer := timer + 1;
    end
  watching CLOSE
    timer := 0;
  end
end
end module
```

2 Module de Sortie Générique

```
module Exit(sensor_id : integer);
input SENSOR_TRIGGER, OPEN, CLOSE, TICK;
output PERSON_EXITED;
integer count := 0;
const integer K = 10; // P riode pour laisser sortir une personne
integer timer := 0;

loop
  await SENSOR_TRIGGER;
  count := count + 1;
  every TICK when count > 0 and present(OPEN) do
    if timer == K then
      emit PERSON_EXITED;
      count := count - 1;
      timer := 0;
    else
```

```

        timer := timer + 1;
    end
    watching CLOSE
        timer := 0;
    end
end
end module

```

3 Module de Contrôle Principal

```

module MuseumControl;
input PERSON_ENTERED, PERSON_EXITED, ALARM, RESET;
output OPEN_ENTRIES, CLOSE_ENTRIES, OPEN_EXITS, CLOSE_EXITS;
const integer N = 100; // Capacit maximale du mus e
integer inside := 0;

loop
    present ALARM then
        emit CLOSE_ENTRIES;
        emit CLOSE_EXITS;
    else present RESET then
        emit OPEN_ENTRIES;
        emit OPEN_EXITS;
    end;
    present PERSON_ENTERED then
        if inside < N then
            inside := inside + 1;
        end;
        if inside >= N then
            emit CLOSE_ENTRIES;
        end
    else present PERSON_EXITED then
        inside := inside - 1;
        if inside < N then
            emit OPEN_ENTRIES;
        end
    end
end
end
end module

```

4 Module pour Calculer la Moyenne de Visiteurs

```

module VisitorAverage;
input PERSON_ENTERED, PERSON_EXITED, TICK;
output AVERAGE_VISITORS;
integer total_visitors := 0;
integer total_time := 0;
float average := 0.0;

loop
  every TICK do
    total_time := total_time + 1;
    if total_time > 0 then
      average := total_visitors / total_time;
      emit AVERAGE_VISITORS(average);
    end;
    watching PERSON_ENTERED do
      total_visitors := total_visitors + 1;
    watching PERSON_EXITED do
      total_visitors := total_visitors - 1;
    end
  end
end module

```

5 Module de Gestion des Entrées pour Fermer une Entrée

```

module EntryManager;
input PERSON_IN_QUEUE, NO_PERSON_IN_QUEUE, TICK;
output CLOSE_ENTRY;
integer open_entries := 3; // Supposons qu'il y a initialement 3 entrées ou
integer queue_count := 0; // Compteur global pour toutes les entrées

loop
  await TICK;
  if queue_count == 0 and open_entries > 2 then
    emit CLOSE_ENTRY;
    open_entries := open_entries - 1;
  end;
  present PERSON_IN_QUEUE do
    queue_count := queue_count + 1;
  watching NO_PERSON_IN_QUEUE do
    queue_count := queue_count - 1;
  end
end module

```